Statement E.71 (Proved). The complete matrix distribution fibres on the full retained word domain. *Put $N = N_B$, $K_6 = \overline{N'N^6}$ and*

$$J = K_6 \cap \widehat{F}', \qquad \widehat{F} = \widehat{F}(x, y). \tag{E.71.1}$$

Keep exactly the 24 *indexed E.17 origins f_i used in E.62 and E.68. Their entire word fibres with the same G_6 value and zero abelianization are $f_i J$.*

For $n = 2, 3$, let δ_n be the positive cyclic-cover Schreier homomorphism of D.5, and put $c_n = \exp_y \circ \delta_n$. Evaluate

$$q\delta_n(f) = q(y)^{c_n(f)} q(f) \tag{E.71.2}$$

in every literal epimorphism to $\mathrm{GL}_2(\mathbf{F}_3)$ and $\mathrm{SL}_2(\mathbf{F}_3)$, represented by the 18 *and* 16 *exact epikernel representatives of E.62 and E.59. Retain c_2, c_3 modulo* 24*, as well as all the finite E.68 conditions and their coefficients modulo six.*

The actual common correction map $\rho : J \to P$, constructed below, has

$$\begin{aligned} |P| &= 2^{105}3^{17}, \quad & |P_0| &= 2^{103}3^{17}, \\ P_0 &= \rho(N'), \quad & P/P_0 &\simeq \mathbf{Z}/4\mathbf{Z}. \end{aligned} \tag{E.71.3}$$

Its old projection is exactly the E.68 image $O = Z \times \mathcal{M} \times \mathcal{G}$, of order $2^{42}3^{13}$. The kernel $P \to O$ has order $2^{63}3^4$.

For each of the 48 *E.68 values over one origin, and for each of the* 16 *lifts of its pair (c_2, c_3) from modulo six to modulo* 24*, there is exactly one value in P satisfying all* 70 *matrix and coefficient conditions (E.71.2). Consequently the complete finite locus has* 768 *values over each indexed origin and* 18 432 *in total. All these values have saved actual word preimages in J. For a saved correction u, the entire corresponding word fibre is*

$$f_i u \ker(\rho). \tag{E.71.4}$$

The retained N' intermediate has exactly four coefficient lifts over each E.68 value, or 4608 *values in total. The cyclic quotient in (E.71.3) is the actual map*

$$[u] \longmapsto c_2(u)/6 \pmod 4. \tag{E.71.5}$$

These assertions determine the displayed finite conditions on the whole domain $f_i J$. They do not assign a full-GT lift or generation in an additional source to any of the saved words.

Proof. We construct the finite actions, identify the whole correction domain, and then determine its fibres.

The exact distribution maps. Let H_n be the kernel of $\widehat{F} \to \mathbf{Z}/n\mathbf{Z}$ sending x to 1 and y to 0. Its positive Schreier generators are x^n and $x^j y x^{-j}$, $0 \le j < n$. Define

$$\delta_n(x^n) = x, \qquad \delta_n(y) = y, \qquad \delta_n(x^j y x^{-j}) = 1 \quad (0 < j < n). \tag{E.71.6}$$

These assignments give an actual continuous homomorphism on H_n. Enriquez's Theorem 8.1, equation (50), and Remark 8.6 prove the corresponding full profinite GT identities and $c_{mn}(f) = c_m(f) + c_n(f)$ [1]; the sign and the comparison with the other Schreier convention are those of D.5.

Write r_n for the least nonnegative residue. For $w \in \widehat{F}$, put

$$j_w = r_n(j + \exp_x(w)), \qquad s_j(w) = x^j w x^{-j_w}. \tag{E.71.7}$$

The index is reduced before it is used as a section exponent. Thus the wrap under x contributes x^n. For a literal epimorphism $q : \widehat{F} \to H$, the following is a right action on $\{0, \dots, n-1\} \times H$:

$$(j, z)w = \big(j_w, q\delta_n(s_j(w))^{-1} z q(w)\big). \tag{E.71.8}$$

Indeed $s_j(ww') = s_j(w)s_{j_w}(w')$; applying the two actions gives the inverse of that product on the left and $q(ww')$ on the right. This proves the morphism, including negative letters and all closed profinite words. On $\widehat{F}'$ the coordinate $j = 0$ is stable, and

$$(0, q(y)^a)f = (0, 1) \quad \Longleftrightarrow \quad q\delta_n(f) = q(y)^a q(f). \tag{E.71.9}$$

An additional right action on $\{0, \dots, n-1\} \times \mathbf{Z}/24\mathbf{Z}$ adds $c_n(s_j(w))$ in its second coordinate. Its condition $(0, -a)f = (0, 0)$ is exactly $c_n(f) = a \pmod{24}$.

The 34 representative maps retain the literal matrix groups and their full automorphism orbits. An equality for a representative transports through its recorded automorphism to every epimorphism in that orbit. The least common multiple of their $q(y)$ orders is 24. There are therefore exactly $2(34+1) = 70$ specified target points. Reduction of the coefficient action modulo six agrees with the E.66 action on each generator in (E.71.6), hence on the entire cyclic-cover group. The other retained coefficients are exactly

$$(c_2, c_3, c_4, c_6, c_{12}) = (a, b, 2a, a+b, 2a+b) \pmod 6. \tag{E.71.10}$$

The new full action has 6360 points. Adjoining the unchanged 5331-point E.68 source gives one actual 11 691-point source. The code first compares the same 97 original words and all the literal matrix tables. No coordinate-zero restriction is made before the required subgroup is identified.

The entire closed word fibre. Let $n_1, \dots, n_{97}$ be the retained free basis of N. Their actual two integral free-source exponent rows give

$$e : \widehat{\mathbf{Z}}^{97} = N^{\mathrm{ab}} \longrightarrow \widehat{\mathbf{Z}}^2. \tag{E.71.11}$$

The saved integer column operations, with their integer inverse checked in both orders, produce $U \in \mathrm{GL}_{97}(\mathbf{Z})$ with

$$eU = \begin{pmatrix} 2 & 0 & 0 & \cdots & 0 \\ 4 & 6 & 0 & \cdots & 0 \end{pmatrix}. \tag{E.71.12}$$

The image index is 12, and its cokernel Smith factors are $2, 6$. They are retained, rather than replaced by a surjective exponent map. Multiplication by every nonzero integer is injective on $\widehat{\mathbf{Z}} = \prod_p \mathbf{Z}_p$. The first two columns in (E.71.12) are therefore injective, so the last 95 columns are exactly $\ker e$ at all primes.

The abelian quotient map identifies K_6/N' with $6\widehat{\mathbf{Z}}^{97}$. It induces the exact surjection

$$J \longrightarrow 6 \ker e, \qquad J/N' = 6\widehat{\mathbf{Z}}^{97} \cap \ker e = 6 \ker e. \tag{E.71.13}$$

For the last equality, $6e(v) = 0$ implies $e(v) = 0$. For surjectivity, lift any $a \in 6 \ker e$ through $N \to N^{\mathrm{ab}}$. Its lift has zero free-source exponents and has abelianization divisible by six, so it lies in J. This is an actual lift through the closed profinite quotient.

For each last column of U, form the finite word $g_j = \prod_{i=1}^{97} n_i^{U_{i,j+2}}$ in increasing generator order. Its two integer exponents are zero. The closed subgroup generated by N' and all the g_j^6 contains the kernel in (E.71.13) and maps onto its entire target; hence it is J. All 95 literal words, columns, inverses and sixth-power values are retained and replayed. Thus their values, together with the complete N' image, generate the entire correction image in every finite source.

The complete finite image and its cyclic quotient. The complete Schreier calculation first determines the image of N on all 11 691 points and then its derived subgroup. This is the actual image of N'. On $J \subset \widehat{F}'$ each coordinate-zero block is stable. Retaining all old 4957 points and the new 2544 coordinate-zero points gives the homomorphism ρ of degree 7501.

Adjoin the 95 actual sixth powers to the computed N' image. Complete presentations and independent word replays give (E.71.3), and show that the old projection remains exactly O. In particular this enlargement has index four. Its complete coset multiplication table, also retained, identifies the quotient as cyclic.

There is a direct coefficient identification of that quotient. Equation (E.71.12) shows $N \subset H_2$, so $c_2|_N$ is a genuine homomorphism and kills N'. Its saved integral row on the last 95 columns has greatest common divisor one. It follows through (E.71.13) that $c_2(J) = 6\widehat{\mathbf{Z}}$. The first kernel word is

$$\begin{aligned} g_0 &= n_1^{-1} n_2 n_3^{-1} \\ &= y^{-5}x^{-1}yx^{-2}yx^{-1}y^2xy^{-1}x^2y^2x, \qquad c_2(g_0) = -3. \end{aligned} \tag{E.71.14}$$

Thus $c_2(g_0^6) = 6 \pmod{24}$. The retained coefficient action makes (E.71.5) well-defined on P; it is onto and kills P_0. Since $[P : P_0] = 4$, its kernel is exactly P_0. This proves the asserted quotient map, including its integral division by six.

Exact fixed-value fibres. The prescribed-base Schreier driver changes no group algorithm: it installs a specified base prefix before running the unchanged complete engine. First use a complete base of O, of length 31. The corresponding stabilizer is precisely $K = \ker(P \to O)$, and the complete certificate retains all its strong generators as words in the original source. Install the 70 target points of (E.71.9) and its coefficient companions as a base for K. The orbit-length product is

$$|K| = 2^{63}3^4, \qquad \mathrm{Stab}_K(\text{all 70 targets}) = 1. \tag{E.71.15}$$

Every final Schreier relation and every strong word is independently replayed. Thus this is the entire kernel, with a trivial simultaneous stabilizer.

For completeness the program rechecks all $24 \cdot 4 \cdot 36 \cdot 4 = 13\,824$ old candidates. Exactly the 1152 E.68 values occur in O. For an old value v, the old-base transversals give an actual preimage $u_0 \in P$. Its full inverse image is u_0K. For each lift (A, B) modulo 24 of the old pair (a, b), apply $f_i u_0$ to the 70 points

$$(0, q(y)^A),\ (0, -A), \qquad (0, q(y)^B),\ (0, -B), \tag{E.71.16}$$

with their respective matrix and coefficient blocks. The remaining problem is the literal simultaneous transporter in K from this tuple to the target tuple. The complete orbit tables decide it, retaining each inverse transversal as an actual kernel word. All $1152 \cdot 16 = 18\,432$ transporters exist. Equation (E.71.15) proves uniqueness of their finite values. The separate P_0 calculation gives exactly four of them: $A = c_2(f_i) \pmod{24}$ and all four compatible values of B. This retains the smaller intermediate without using it to exclude a full fibre.

Finally substitute the kernel and old-base transversals into their source certificates, then substitute the 95 lattice words. The resulting DAG is over the original 97 words. All 18 432 values are replayed on the full 11 691 points. Every saved correction has zero exact integral free-source exponents and zero N-abelian exponents modulo six; hence it belongs to J. Its retained value is the one just counted. Equality under the homomorphism ρ is exactly the kernel coset (E.71.4).

The programs use the prefix `a_cm_kn23_`: `model`, `source`, `ab_lattice`, `full`, `fibres`, `witnesses`, `nprime_comparison`, `coefficient` and `numeric_summary`. Their outputs retain the actual matrices, integral lattice, complete presentations, transporter paths and all common word preimages. The unsuccessful preliminary elementary-abelian guess for the index-four quotient is separately preserved; its corrected exact multiplication table and (E.71.5) give the cyclic answer used here. □

References

[1] B. Enriquez, Quasi-reflection algebras and cyclotomic associators, Selecta Mathematica 13 (2007), 391–463.